AIML-401: Deep Learning — Revision Study Guide

Topic: **Train Split**

Candidate ID: STU-807	**Target Exam:** Midterm IA-2 & SEE Exam
Curriculum Unit: Unit 1: Mathematical Foundations	**Compiled:** Nov 02, 2026 - 21:53
Engine: Autonomous Python Script Execution	**Platform:** PACD-RAG Academic Intelligence

1. Fundamental Concept & Definition

Train Split is a key academic methodology in machine learning and deep learning pipelines.

2. Executed Python Code Implementation

```
from sklearn.model_selection import train_test_split
import numpy as np
X = np.random.randn(100, 4)
y = np.random.randint(0, 2, size=100)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42
, stratify=y)
print(f'Dataset: {X.shape} -> Train: {X_train.shape}, Test: {X_test.shape}')
```

Runtime Execution Output: *Dataset: (100, 4) -> Train: (80, 4), Test: (20, 4)*

3. 4-Phase Execution Pipeline

Phase	Name	Mathematical / Code Operation	Key Objective
1	Ingestion	Load dataset matrices X and y	Input validation
2	Partitioning	Stratified subset separation	Preserve priors
3	Isolation	D_train intersect D_test = empty set	Prevent leakage
4	Evaluation	Fit on train, score on test	Unbiased metric

4. High-Yield Question & Pitfall Checklist for Exams

Top Exam Trap: Applying preprocessing normalization before dataset partitioning leads to subtle data leakage.
Expected Question (8 Marks): *'Explain the core mechanisms, mathematical formulation, and pitfalls of Train Split. Detail the prevention of data leakage.'*
Marking Rubric:
• 2 Marks: Theoretical formulation
• 3 Marks: Implementation / Execution pipeline
• 3 Marks: Pitfall avoidance